

approximately uniform after root mean square normalization, leaving directional variation as the primary source of information [17, 51, 52]. For example, in meta-llama/Llama-3.1-8B, the mean norm is around 18.8 across four different datasets; see Figure 5 in the appendix for details. We also observe that $f_\theta(x) \mid y$ exhibits unimodal structure (see Figure 6 in the appendix), which motivates the use of the vMF model in deriving (2).

Representation Steering. Although the base extractor $f_\theta(\cdot)$ captures high-level semantic information, the resulting class-conditional distributions may still overlap in latent space, even in out-of-distribution settings (see Appendix F.1 for evidence).

To improve separability, we introduce a learnable steering vector $\mathbf{v} \in \mathbb{R}^d$ and define a steered extractor $f_{\theta,\mathbf{v}}(\cdot)$. We implement steering as an additive intervention on intermediate activations of the observer model by injecting a vector $\mathbf{v}$ at a chosen layer ℓ :

$$h_{\ell,t}(\cdot) \leftarrow h_{\ell,t}(\cdot) + \mathbf{v}.$$

In the default setting, the intervention is applied at an intermediate layer $\ell \leq L - N + 1$. Importantly, the method is not sensitive to the exact choice of ℓ : as long as the intervention is applied within intermediate layers, performance remains stable (see Section 4.2).

This is the only modification to the model: after the injection, the observer model remains fixed, and the perturbed activations propagate through subsequent Transformer blocks without further intervention, producing a new representation. Intuitively, since the model captures rich semantic structure from large-scale pre-training, the steering vector $\mathbf{v}$ introduces a direction that enhances its intrinsic discriminative capability. Applying the same token selection, layer aggregation, and normalization yields the steered representation $f_{\theta,\mathbf{v}}(\cdot) \in \mathbb{S}^{d-1}$ for the downstream hypothesis test.

2.3 Phase I: Learning the Steering Vector via Likelihood Maximization

Likelihood-Based Modeling. A remaining question is how to learn the steering vector $\mathbf{v}$, which is addressed in Phase I of our method. The steering vector aims to enhance the separability of latent representations across classes. To quantify this, we formulate a classification problem where the feature is given by $f_{\theta,\mathbf{v}}$ and the response is the ground-truth label ($y_i = 1$ for LLM-generated text and $y_i = 0$ for human-written text). More specifically, we model the conditional distribution $f_{\theta,\mathbf{v}}(x_i) \mid y_i$ as a von Mises–Fisher (vMF) distribution, a directional analogue of a Gaussian on the unit sphere.² Under a shared-concentration vMF model, the class posterior takes a softmax form. Motivated by this structure, we use the same functional form for Phase-I optimization, with the inverse temperature $\beta > 0$. Specifically, under a uniform class prior, we define

$$q_\beta(y_i \mid f_{\theta,\mathbf{v}}(x_i), \boldsymbol{\mu}_0, \boldsymbol{\mu}_1) = \frac{\exp(\beta \boldsymbol{\mu}_{y_i}^\top f_{\theta,\mathbf{v}}(x_i))}{\sum_{c \in \{0,1\}} \exp(\beta \boldsymbol{\mu}_c^\top f_{\theta,\mathbf{v}}(x_i))}. \quad (2)$$

Here, $\boldsymbol{\mu}_c \in \mathbb{S}^{d-1}$ denotes the mean direction for class $c \in \{0,1\}$, while β is an optimization hyperparameter and is distinct from the shared vMF concentration parameter κ . Appendix B derives the corresponding softmax posterior under the vMF model and explains this temperature-scaled formulation. To estimate the steering vector, we collect a labeled training set $\mathcal{S}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{n_1}$ and optimize the following objective:

$$(\hat{\mathbf{v}}, \hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1) = \arg \max_{\mathbf{v} \in \mathbb{R}^d, \boldsymbol{\mu}_0, \boldsymbol{\mu}_1 \in \mathbb{S}^{d-1}} \frac{1}{n_1} \sum_{i=1}^{n_1} \log q_\beta(y_i \mid f_{\theta,\mathbf{v}}(x_i), \boldsymbol{\mu}_0, \boldsymbol{\mu}_1). \quad (3)$$

Optimization Procedure. In practice, we adopt a two-timescale optimization scheme to optimize (3). Specifically, the steering vector $\mathbf{v}$ is updated via gradient ascent on a slower timescale (i.e., using a smaller step size η), while the class mean directions $\boldsymbol{\mu}_c$ are updated on a faster timescale (i.e., using an exponential moving average (EMA) with coefficient $\rho \in (0,1)$) over normalized class-specific embeddings [53]. The observer model remains fixed throughout the optimization, and we choose $0 < \eta \ll \rho \leq 1$ to introduce two time scales. The full procedure is provided in Appendix A due to space constraints.

²Under the vMF model, $p(f_{\theta,\mathbf{v}}(x_i) \mid y_i = c) \propto \exp(\kappa \boldsymbol{\mu}_c^\top f_{\theta,\mathbf{v}}(x_i))$, where κ is the concentration parameter. The Phase-I objective retains the resulting softmax form but uses a separate inverse temperature β .

Remark 2.2 (Gradient analysis). *Intuitively, this objective in (3) encourages the steering vector to reshape the representation space so that class-wise representations become more separable. In Appendix H.1, we characterize the gradient of the population version of (3), which provides theoretical support for this effect.*

2.4 Phase II: Detection via the Log-Likelihood Ratio Test

Log-Likelihood Ratio Test. Detection naturally reduces to a log-likelihood ratio test. Under the shared-concentration vMF generative model, the Neyman-Pearson (NP) lemma [54] implies that the most powerful test is based on the log-likelihood ratio:

$$\mathcal{S}(x) = \log \left(\frac{p(f_{\theta, \mathbf{v}}(x) \mid y = 1, \boldsymbol{\mu}_1, \boldsymbol{\mu}_0)}{p(f_{\theta, \mathbf{v}}(x) \mid y = 0, \boldsymbol{\mu}_1, \boldsymbol{\mu}_0)} \right) = \kappa(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)^\top f_{\theta, \mathbf{v}}(x). \quad (4)$$

Since $\kappa > 0$ only rescales the score, it does not affect the rejection region after threshold calibration. Thus, the detector is equivalently based on projecting the steered representation onto the discriminative direction $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0$, with unscaled projection used in practice. In practice, we plug in the estimates $(\hat{\mathbf{v}}, \hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1)$ obtained from (3) to form the empirical score $\hat{\mathcal{S}}(x)$. The resulting detector is $\hat{\phi}(x) := \mathbb{1}(\hat{\mathcal{S}}(x) > \hat{\tau})$, which rejects the null hypothesis $\mathcal{H}_0$ when the score exceeds the threshold $\hat{\tau}$. Accordingly, x is classified as LLM-generated if $\hat{\phi}(x) = 1$, and as human-written otherwise.

Threshold Calibration. In high-stakes scenarios such as academic integrity assessment, falsely labeling human-written text as LLM-generated corresponds to a Type-I error with substantial practical and ethical consequences. In such settings, controlling the false positive rate (FPR) is more important than optimizing a balanced trade-off between detection power and false alarms. Accordingly, we calibrate the threshold using an independent calibration sample $\mathcal{S}_{\text{cal}} = \{x_i^-\}_{i=1}^{n_2}$ drawn from the null distribution, and define

$$\hat{\tau}_\alpha := \inf \left\{ \tau : \frac{1}{n_2} \sum_{i=1}^{n_2} \mathbb{1}(\hat{\mathcal{S}}(x_i^-) > \tau) \leq \alpha \right\}. \quad (5)$$

This empirical calibration enforces the target Type-I error level on the calibration set. Moreover, Theorem 3.1 shows that the resulting test achieves approximate control of the population Type-I error in finite samples with high probability.

Remark 2.3 (Threshold from Youden index). *In Appendix C, we describe an alternative threshold selection strategy based on the Youden index, which is commonly used in standard benchmarking settings [33, 55–57]. While this approach is not designed to control the Type I error, our empirical results indicate that it can nevertheless achieve competitive false positive control in practice.*

3 Theoretical Analysis

In this section, we provide a theoretical analysis of our method. We begin by establishing finite-sample guarantees for Type I error control, and then characterize the excess Type II error arising from parameter estimation. Finally, we study the effect of null distribution shifts on the resulting test. All proofs are deferred to Appendix G.

Type-I Error Control. We show that empirical threshold calibration ensures Type I error control with high probability. Let $\hat{\mathcal{S}}_t$ denote the score function at training iteration t , trained on $\mathcal{S}_{\text{train}}$, and define the detector $\hat{\phi}(x) := \mathbb{1}(\hat{\mathcal{S}}(x) > \hat{\tau}_\alpha)$, where $\hat{\tau}_{\alpha, t}$ is the empirical threshold at iteration t , computed from an independent null sample $\mathcal{S}_{\text{cal}} = \{x_i^-\}_{i=1}^{n_2}$ as in (5). Theorem 3.1 shows that this procedure controls the Type I error up to a finite-sample deviation that vanishes as n_2 increases.

Theorem 3.1 (Type-I Error Control). *Let $\alpha, \delta \in (0, 1)$ and suppose that, conditional on the training sample, the null score $\hat{\mathcal{S}}_t(X^-)$ has a continuous distribution. Then, with probability at least $1 - \delta$ over the independent calibration sample, $\left| \mathbb{P}_0 \left(\hat{\phi}(X_{\text{test}}^-) = 1 \right) - \alpha \right| \leq \sqrt{\log(2/\delta)/(2n_2)} + 1/n_2$.*

Remark 3.2 (Ties). *The continuity assumption ensures almost surely distinct calibration scores. With ties, one may conservatively exclude the boundary group for one-sided Type-I control or use randomized tie-breaking for exact empirical calibration.*

Excess Type-II Error. Next, we study the Type-II error. In general, analyzing the power of the detector is challenging due to the complex and implicit behavior of LLM-extracted representations. To facilitate analysis, we consider a simplified setting in which the extracted features $f_{\theta, \mathbf{v}}(X)$ follow a class-conditional vMF model with certain ground-truth parameters μ_0 and μ_1 . This assumption allows us to define the corresponding oracle likelihood ratio test (i.e., $\mathcal{S}$ in (4)) based on this true model, and to study the excess Type-II error of the plug-in detector $\hat{\mathcal{S}}$ relative to this oracle benchmark. In particular, it enables us to disentangle the contributions of estimation and calibration.

Importantly, the oracle and plug-in scores are evaluated at the same steering iterate $\mathbf{v}_t$ and operate on the same steered representation $f_{\theta, \mathbf{v}_t}(x)$. The following result therefore isolates prototype-estimation and threshold-calibration errors, rather than comparing $\mathbf{v}_t$ with a globally optimal steering vector or establishing global convergence of the steering optimization.

Theorem 3.3 (Excess Type-II Error (informal)). *Assume the extracted features $f_{\theta, \mathbf{v}}$ follow a vMF distribution, and the EMA coefficient ρ and learning rate η are chosen sufficiently small, satisfying $0 < \eta \ll \rho < 1$. Under regularity conditions, there exists a constant $c \in (0, 1)$ such that, with probability at least $1 - 2\delta$, for any fixed $0 \leq t \leq T$, the gap $\mathbb{P}_1(\hat{\phi}(X_{\text{test}}^+) = 0) - \mathbb{P}_1(\phi(X_{\text{test}}^+) = 0)$ is bounded by*

$$\mathcal{O}\left(\underbrace{\left((1 - c\rho)^t + \sqrt{\rho \log \frac{4T}{\delta}} + \frac{\eta^2}{\rho^2}\right)^{\frac{1+\bar{\gamma}}{2}}}_{\text{Estimation Error}} + \underbrace{\sqrt{\frac{\log(2/\delta)}{n_2}} + \frac{1}{n_2}}_{\text{Calibration Error}}\right)$$

where n_2 is the calibration sample size and $\bar{\gamma}$ characterizes the local concentration of the score distribution (i.e., $\mathcal{S}$ under $\mathbb{P}_0$) around the threshold τ_α^* .³

Theorem 3.3 establishes an upper bound on the excess Type-II error, which characterizes the sub-optimality of the learned score function $\hat{\mathcal{S}}$. The bound consists of two components: an estimation error arising from learning the ground-truth parameters μ_0 and μ_1 , and a calibration error due to estimating the critical value $\hat{\tau}_{\alpha, t}$.

The second term follows from Theorem 3.1 and reflects the finite-sample effect of threshold calibration; it decreases as the calibration sample size n_2 increases. The first term can be further decomposed into several parts: an optimization term exhibiting geometric decay $(1 - c\rho)^t$, a variance term induced by stochastic training $\sqrt{\rho \log(4T/\delta)}$, and a higher-order lag term η^2/ρ^2 . When t is sufficiently large and the two-timescale condition $\eta \ll \rho < 1$ holds with ρ sufficiently small, the estimation error becomes negligible.

Effects of Distributional Shift. Prior work has highlighted performance degradation under domain shifts in LLM-generated text detection [58, 59], and proposed database-based strategies to mitigate such a mismatch. However, these approaches do not explicitly characterize the impact of distribution shift. In Appendix H.2, we quantify this effect by analyzing the detector under a Wasserstein perturbation with magnitude $\mathcal{E}$. Proposition H.2 provides finite-sample bounds for both Type-I and Type-II errors under the shifted distributions. In particular, when $\mathcal{E} > 0$, both errors may deteriorate, reflecting the impact of distribution mismatch.

4 Experiment

In this section, we present a comprehensive evaluation of the proposed S2D detector. Our code is available at: <https://github.com/LuxLiang/steer-to-detect-release/tree/main>.

Datasets. Following previous setups [33, 60], we evaluate S2D on the DetectRL benchmark [61], which covers four domains that are particularly vulnerable to misuse by LLMs, including academic writing (arXiv Archive⁴), news summarization (XSum [62]), creative writing (WritingPrompts [63]), and user reviews (Yelp [64]). For each domain, the benchmark provides 2,800 pairs of human-written and LLM-generated texts. The LLM-generated texts are produced by four widely used LLMs: *GPT-3.5-Turbo* [65], *Claude-Instant* [66], *Google-PaLM* [67], and *Llama-2-70B* [68]. To ensure robust

³A larger value of $\bar{\gamma}$ corresponds to less mass near the threshold and hence an easier estimation problem. See Assumption 3 for the detailed definition.

⁴<https://www.kaggle.com/datasets/spsayakpaul/arxiv-paper-abstracts/data>.